

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-4;

AGRICULTURE-2

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2. CAPITAL FORMATION IN AGRICULTURE SECTOR

- **Practice Question:** Why is it important to increase the gross capital formation (GCF) in agriculture sector? Discuss the key steps taken by government towards achieving this. What are the key factors hindering the increase in GCF in the sector? [15 marks, 250 words]
- **Gross Capital Formation (GCF)** refers to the total investment in physical assets over a specific period. It includes new and existing physical assets, such as machinery, buildings, land improvements, equipment purchases, inventory changes, investment in irrigation facilities etc.
- **Why capital formation is crucial?**
 - Modernization Agriculture
 - Enhancing Productivity
 - **Increasing farmer's income**
 - **Enhancing economic growth in rural India**
 - **Enhancing sustainability of agriculture sector** (e.g. through micro-irrigation tools, investment in precision agriculture etc)
- **Various government initiatives to increase capital formation in agriculture sector:**
 - **The Agriculture Marketing Infrastructure (AMI) Sub-scheme** of the Integrated Scheme for Agriculture Marketing (ISAM) was launched in 2014. It is aimed at improving the extent of storage infrastructure through demand driven, **credit linked scheme offering subsidies of 25% (for the plains) and 33.33% (for NE and Hilly regions)** to individuals, farmers, FPOs, cooperatives and state agencies.
 - As of 30th April 2024, more than 48,000 projects were sanctioned for storage infrastructure with Rs 4570 crore released as subsidy.
 - **Agriculture Infrastructure Fund (AIF)** was launched with a financing facility of Rs 1 lakh crores to be disbursed between FY21 to FY26 with support extending till FY33.
 - It provides medium-term debt finance for **post-harvest management and community farming projects**, offering interest subvention and credit guarantee support.
 - As of July 2024, AIF has mobilized an investment of Rs 73194 crores, supporting 17K+ CHCs, 14K+ primary processing units etc.
 - **Pradhan Mantri Krishi Sampada Yojana** has also introduced credit-linked financial assistance through grants in aid to build efficient supply chain management from farm to retail to **reduce wastage of perishable produce** and extend food shelf life.
 - Under PMKSY more than 1600 projects have been completed by March 2024.
 - **The three farm laws** (which were eventually repealed in 2021) were also focused on increasing the investment of private sector in agriculture.
- **What is the situation of GCF in agriculture sector?**
 - The GCF of agriculture sector and the share of GCF in the agriculture and allied sectors as a percentage of Gross Value Added (GVA) has been growing steadily, mainly due to increased public investment.
 - » The GCF of agriculture sector grew at the rate of 19.04 percent in 2022-23
 - » The GCF as a percentage of GVA rose from 17.7% in 2021-22 to 19.9% in 2022-23, **suggesting an increase in investment in agriculture sector.**
 - The average annual growth in GCF from 2016-17 to 2022-23 was 9.70%.

- **Need to further increase GCF:**
 - » The DFI 2016 report indicated that to double farmer's income over 2016-17 to 2022-23, income would need to grow at an annual rate of 10.4% in the farm sector, which would in turn require an annual growth rate in agriculture investment of 12.5%.
- **Key challenges hindering investment in agriculture sector:**
 - » Fragmentation of Agricultural land
 - » Poor participation of private corporate sector whose share has remained below 2%.
 - » Public investment has grown, but subsidy bill takes a lot of of government's spending on agriculture.
 - Public investment remained at about 1/3rd of the subsidies.
 - » **Low farm incomes** discourage farmers to go for large scale investment in the sector.
- **Conclusion:**
 - An integrated and holistic approach consisting of increased subsidy on machinery and CHCs, consolidation of landholding, promoting private investment and rationalization of subsidy expenses will be needed to enhance the gross capital formation in agriculture sector in India.

3. AGRI-CREDIT

1) QUESTION-1

- What is Agri-Credit, and how does it support the agriculture sector in India? Discuss the key steps taken over the years to improve agri-credit situation in the country [10 marks, 150 words]
- **Introduction**

Definition Based: Agri-Credit is the credit designed to meet the financing needs of agricultural farmers and businesses. It enables farmers and agricultural enterprises to access funds for agri-production, equipment purchase, expanding operations, warehousing etc.

Data - Based: Agriculture credit as a percentage of Agriculture GDP increased from 2% percent in 1970s to 47% by 2019-20, portraying significant progress made in lending to agriculture.
- **Steps Taken to Promote Agri-Credit over the years**
 - » **Nationalization of Commercial Banks** in 1969, Establishment of Regional Rural Banks in 1976 and setting up of NABARD in 1982 have been some of the biggest steps towards increasing farm credit.
 - » **Priority Sector Lending (PSL)** norms initiated in 1974, mandates all domestic commercial banks (and foreign banks with 20 or more branches) to earmark 18% of loans for farm credit.
 - » **Kisan Credit Card (KCC)** introduced in 1998 are aimed at providing adequate and timely short term credit needs for farmers and has been extended to fishery and animal husbandry activities since 2018-19. Further, the limit for collateral free loan has been expanded to 1.6 lakh crores.
 - » **Modified Interest Subvention Scheme**, operational since 2006-07, provides short term agri credit of upto Rs 3 lakh at a subsidized interest rate of 7% per annum to farmers engaged in agriculture and allied activities. An **additional 3% subvention (Prompt repayment incentive)** is also given to farmers for prompt and timely repayment of loans.
 - » **Joint Liability Groups Scheme** was initiated by NABARD in 2006.

- To bring small, marginal, tenant farmers, oral lessees, etc. taking up farm activities, off-farm activities and non-farm activities, into the fold of institutional credit, Joint Liability Groups (JLGs) have been promoted by banks.
- As per ESI 2023-24, JLG accounts have grown at a compound annual growth rate (CAGR) of 43.76% over the past five years, emerging as a vital source in meeting the credit needs tenant farmers and marginalized segments.
- » **Relief Measures during natural calamities:** This include restructuring/rescheduling of existing crop loans and term loans, extending fresh loans, relaxed security and margin norms, moratorium etc.
- » **Agriculture Infrastructure Fund** announced as a part of AtmaNirbhar Bharat Abhiyan will further boost the credit flow to agri-sector.

- **Impact of the above initiatives ->**

- » **Spectacular growth in agri credit sector** (from 2% of agri-GDP in 1970s to around 40% in 2019-20)
- » **Enhanced agri productivity -> Food Security**
- » **Increased investment** in agriculture infrastructure such as irrigation systems, storage facilities etc.
- » **Diversification and Modernization** -> Easy credit can give farmers scope to new crops, adopt organic farming or invest in livestock sector
- » **Income Stability for farmers:** Easy farm credit can reduce dependency on unauthorized money lenders and allow them to meet their expenses through institutional credit
- » **Increased scope for entrepreneurship, agribusiness development** -> Rural employment and poverty alleviation
- » **Increased contribution of commercial banks:** In India, scheduled commercial banks (79%) are the major players in supplying agri credit followed by rural cooperative banks (15%), RRBs (5%) and MFIs (1%).

- **Conclusion:** Class Discussion

2) QUES-2:

- **Introduction:**

- **As per RBI's Internal Working Group on Agri Credit,** the system suffers from following limitations:

- » **Skewed Regional Distribution of Agricultural Credit in India:** It is skewed in favour of the southern states (40%). The share of **north-eastern states (2%)** has been very low.
 - **Why low coverage for agri-credit in NER:** Total cultivable area in North-east is only 2.75% of the total GCA of the country. Prevalence of Community Ownership in most of NE stats -> affects KCC loans since they are provided against land records;
- » **Mostly benefit large and medium farmers** (Small and marginal farmers, sharecroppers, tenants etc., who are highly vulnerable find it difficult to access it). Small farmers are **still dependent on non-institutional credit** which is available at higher rates
- » **Biased towards crops** and allied activities like animal husbandry, fishery etc are ignored.
 - Allied activities, In spite of providing around 40% of agri-output get around 6-7% of the farm credit.
- » **Low Capital Formation:** Most of the agri credit is short term in nature and is being used for production purposes without any substantial improvement to agri investment.
- » **Loan waivers negatively impact credit culture and weaken state finances:** It harms farmers' interest in medium and long term and also squeeze the fiscal space of governments to increase the productive investment in agricultural infrastructure.

- **Way Forward: How to deal with above challenges**
 - » **Improving Agri-Credit Situation:**
 - Promote more long-term agri-infrastructure loans.
 - Set up a Credit Guarantee Fund on lines of one being implemented for MSME sector.
 - Extend the limit of collateral free loans to 5 lakhs.
 - » **Synergize Basket of Financial Initiatives with use of Technology**
 - States should fast track the digitization and Updation of land records in a time bound manner. This will fast track and increase the scope of farm credit.
 - » **Reduce the skewedness in agri-credit sector**
 - Under PSL, increase the share for small and marginal farmers
 - Bring in the provision of combo loans (loans for agri activities + loans for allied sector)
 - States should **reform restrictive land leasing laws**
 - On the basis of Model Land Leasing Act proposed by NITI Aayog.
 - This will help tenants to get credit and invest in land.
 - » **Facilitate and promote long term loans to farmers** -> this will contribute to improved agriculture infrastructure.
 - » **Avoid Blanket Loan Waivers** -> creates moral hazards and is harmful in long run
 - » **Enhance focus on public investment:** Increased Gross Capital formation through public investment will stimulate demand for investment credit in agriculture sector.
 - » **Cooperative Federalism for agri Reform:**
 - Set **up a federal institution like GST Council for effective Agri-Reforms** through cooperative federalism.
 - This kind of body will increase consultation with states and bring them on board on various agri-reforms. This will take forward cooperative federalism in agri-sector.
- **Conclusion**
 - » While various reforms over the last 50 years have improved farm credit situation in the country, more work is needed to make the system more inclusive, equitable and effective.

4. IRRIGATION AND ASSOCIATED ISSUES

1) IRRIGATION SITUATION IN INDIA

- **Questions:**
 - » How does irrigation facilities impact Agri-Production in India? Discuss various types of irrigation systems used in India. Analyze their merits, demerits and suitability in different geographical regions [15 marks, 250 words]
 - » Discuss some key government initiatives for increasing the coverage of irrigation in India. Suggest some innovative and sustainable solutions to improve irrigation efficiency [15 marks, 250 words]
- **Introduction**
 - » Irrigation is crucial to ensure that farmers reap full benefit of better-quality seeds and fertilizers. But, as per ESI 2023-24, only 55% of India's total cropped area is irrigated. Rest depends on

Monsoon rainfall for agriculture. This is the most important factor which makes farming a vulnerable profession.

- **Importance of Irrigation:**

- » **Insufficient, Uncertain and Irregular Rains:** In India rainfall is restricted to four months of Monsoon. Even in this period, the global warming and climate change is increasing variability. In this situation, irrigation facilities can crucially control droughts and famines.
- » **Higher Productivity of Irrigated Fields:** Several studies have shown that both productivity (per hectare yield of food grains) and growth in yield are both higher in case of irrigated land.
- » **Multiple Cropping Possible:** Since, India has tropical and subtropical climate in most of the country, there is a potential to grow crops all year around if enough water for irrigation is available.
- » **Brining More Land Under Cultivation:** Possibility of irrigation can make some areas like barren lands, fallow lands etc. under cultivation.
- » **High Yielding Varities Program's success** depends to a large extent on the timely availability of ample supply of water.
- » **Reduces Instability in output levels:** It plays a 'protective role' during drought years. It also plays a role in ensuring stability of income and employment in rural areas.
- » **Other Indirect Benefits of Irrigation:** employment potential of irrigated land increases. It also help in developing allied activities, means of water transport etc.
 - For e.g. it has been estimated that for every Rs 100 of direct benefits from Bhakra Nangal Dam, there was a generation of Rs 90 of indirect benefits.

- **Different types of irrigation in India:**

- » Sources of Irrigation in India can be divided into the following: **(i) Wells/Tube Wells (ii) Canals (iii) Tanks, and Others**
- **Wells and Tube-wells** are the most important source of irrigation in India.
 - » They are spread over large areas of Punjab, Haryana, Uttar Pradesh, Bihar, Rajasthan, and Tamil Nadu.
 - » **Advantages of Well and Tube well Irrigation:** Simplest and cost-effective way, easily affordable by poor Indian farmers (wells). It is an independent source of irrigation and can thus be used whenever the necessity arises. Certain chemicals from ground water such as nitrate, chloride, sulfate etc. are generally found mixed in well water. This can be good for agriculture.
 - » **Disadvantages:**
 - Only limited area can be irrigated (upto 1 to 8 hectare of land per day)
 - Excessive extraction has caused ground water depletion in several parts of the country.
- **Canals** contribute to irrigation of around 24% of the irrigated area. This includes large area of Punjab, Rajasthan, Haryana, Uttar Pradesh, Bihar and some parts of southern states.
 - » Digging of Canals in stony and uneven areas is difficult and unprofitable. Thus, the canals are practically absent from the Peninsular plateau. However, the coastal and delta region of south India do have some canal system.
 - » **Advantages of Canal Irrigation:**
 - Canals can convert dry regions into fertile territory (e.g. Rajasthan impacted by Indira Gandhi Canal)

- They carry a lot of sediments carried by rivers -> this when deposited in agri field contributes to soil fertility.
 - It's quite cheap in long run (initially it can be expensive due to the cost of multipurpose projects etc.)
- » **Drawbacks:**
 - **Water logging** along the canal route: It is caused by soaking on canal water into the ground.
 - **Land Degradation:** Capillary action brings alkaline salts to the surface and makes large areas unfit for agriculture.
 - **Overflow of canals during rainy season** also becomes a reason for floods.
- » Taken together, **Canals and wells** accounted for 86% of the irrigated area in 2012-13
- **Tank Irrigation** is resorted to mostly in Telangana, Andhra Pradesh, Tamil Nadu, and parts of West Bengal and Bihar.
 - » A **tank** act as an irrigation storage system that is developed by constructing a small bund of earth or stones built across a stream. Tanks are of varying size but most of the tanks are of small size and are built by individual farmers or group of farmers.
 - » Rivers of south India don't flow all the year around. Therefore, tanks are constructed for storing water in rainy season which is subsequently used for irrigation purposes.
 - » **Andhra Pradesh** (including Telangana) is the largest state of tank irrigation which has about 29% of tank irrigated area of India. The drainage system of Godavari and its tributaries have a large number of tanks.
 - » **Advantages:**
 - Most of the tanks are natural and not expensive for their construction. Even an individual farmer has his own tank.
 - They are generally constructed on a rocky bed and has long life.
 - Fishing activities in some tanks adds to the food resources and income of the farmers.
 - » **Limitations:**
 - They dry up during dry season and fail to provide the irrigation when it is needed the most.
 - **Silting** of the tank bed is a serious problem and it requires desilting of the tank at regular interval.
- **Micro Irrigation** (including Sprinklers, Drip Irrigation Etc.) have emerged as the new efficient way of irrigation.
- **Various Initiatives** to promote increase irrigation cover in India:
 - » **Accelerated Irrigation Benefit Program (AIBP)** was launched in 1996-97 to provide Central Loan Assistance to states for completion of large and medium irrigation project which have been stuck for long due to fund crunch. It has been now subsumed under PMKSY (PMKSY-AIBP)
 - » **Pradhan Mantri Krishi Sinchai Yojana** (PMKSY) is operational since 2015 with the vision of extending the coverage of irrigation "Har Khet ko Pani" and Improving water use efficiency "More Crop Per Drop".

- » **Promoting water use efficiency:**
 - To promote micro-irrigation a **Micro Irrigation Fund (MIF)** with corpus of Rs 5,000 crore was created with NABARD during 2018-19.
 - Micro irrigation is also being promoted through the **Per Drop More Crop component** of PMKSY (PMKSY-PDMC) from 2015-16.
- » **Watershed Development Program** (now part of PMKSY) also focuses on improving irrigation situation in rain-fed area.
- **Despite the fact that government has spent lakhs of crore of rupees since independence on irrigation programs, it suffers from a number of limitations.** The key reasons are:
 - » **Delays in project Completion:** A large number of major irrigation projects have been pending since last 30-40 years. Not only this, but there is also a delay in utilization because the field channels haven't been created.
 - » **Inter-State Water Disputes:** Generally, most of the major rivers are inter-state. In the absence of effective cooperative federalism, the differences with regard to storage, priorities and use of water arise between different states and impede the development of water resources of some of the rivers.
 - » **Regional Disparities in irrigation development:** Most of the irrigation facility development has taken place in northern region. It is the least in north-eastern region.
 - » **Waterlogging and Salinity:** States like Haryana and Punjab are suffering from waterlogging and salinity (due to capillary action).
 - » **Ageing of Infrastructure and increased Siltation:** Almost 60% of the total dams of the country are more than 3 decades old. Besides regular maintenance, many older structures need replenishment for which funds are a constraint.
 - » **Tail Ender Deprivation:** This is leading to lower agri-productivity for tail-end farmers
 - » **Decline in water Table:** There has been a steady decline in water table in the recent period in several parts of the country, especially in the western India, on account of over-exploitation of ground water and insufficient recharge from rainwater.
 - » **Wastage and Inefficiencies** in water use:
 - Flood irrigation, low cost of electricity is leading to inefficient use of water. Further, in canals also around 44% of water is lost before entering the agricultural field.
- **Way Forward:**
 - » **Improve the implementation of Accelerated Irrigation Benefit Program:**
 - **Improve Work Management:** Appropriate checks; fix accountability for deficiencies; mechanism to monitor the due diligence of states.
 - **Improve Monitoring:** Use of satellite imagery and field report; Strengthen the participatory model of irrigation through water users associations to tackle various issues related to irrigation.
 - **Improved Community Participation:** Farmers need to be involved in the planning and implementation of irrigation projects in order to ensure that they are effective and substantial.
 - » **Promoting Water Efficiency:** Precision Agriculture, Micro-Irrigation are the way towards reducing water dependency of India's agriculture.
 - **Rationalizing Cost of Electricity and water** to promote water use efficiency.

- » **Water Conservation Initiatives** like Jal Shakti Abhiyan etc. needs to be intensified, for long term sustainability
- » **Crop Diversification:** Farmers needs to move towards crops which consume less water (e.g. millets, pulses etc.)
- » **Research and Development:** Develop crop varieties which can survive less water situation.
- » **Farmer's Education and Training:** To ensure farmers are aware of modern farming techniques, crop diversification, climate smart agriculture etc.

3) PRADHAN MANTRI KRISHI SICHAYI YOJNA (PMKSY)

- **Question:** The Pradhan Mantri Krishi Sinchayi Yojna (PMKSY) aims to extend the coverage of irrigation in India. Discuss the key features of this scheme and evaluate its effectiveness [15 marks, 250 words]
- **Introduction:** Pradhan Mantri Krishi Sinchayi Yojna (PMKSY) is a flagship scheme launched by Government of India in 2015. the main objective of the PMKSY is to:
 - » Achieve convergence of investments in irrigation at the field level.
 - » Expand cultivable area under irrigation ("Har Khet ko Pani")
 - » Improve on-farm water use efficiency ("More Crop Per Drop") to reduce wastage of water, Enhance the adoption of precision irrigation and other water saving technologies (Per Drop More Crop)
 - » Enhance recharge of aquifers and introduce **sustainable water conservation** practices by exploring the feasibility of reusing treated municipal wastewater for peri-urban agriculture and attract greater private investment in precision irrigation.
- **Key Features of the Scheme:**
 - » **Amalgamation of Ongoing Schemes:**
 - Accelerated Irrigation Benefit Program (AIBP) of Ministry of Water Resource, River Development & Ganga Rejuvenation (MoWR, RD&GR) (Now Ministry of Jal Shakti)
 - Integrated Watershed Management Program (IWMP) of Department of Land Resources (DoLR) - MoRD
 - On Farm Water Management (OFWM) of Department of Agriculture and Cooperation - Ministry of Agriculture and Farmer Welfare
 - » **Decentralized State-Level Planning and Execution:** States will draw their own irrigation development plan based on district irrigation plans and state irrigation plans.
 - » It serves as a **convergence platform** for all water sector activities including drinking water and sanitation, MGNREGA, application and S&T through comprehensive plan.
 - » **Focused on "Protective Irrigation"** by sustainably water conservation by harnessing rain water at micro level through 'Jal Sanchay' and 'Jal Sinchan'
 - » **Special Focus on Micro-Irrigation** to increase water use efficiency.
 - » **Other objectives** include enhancing recharge of acquifers and introducing sustainable water conservation practice by exploring the feasibility of reusing municipal waste water and peri-urban agriculture and attract greater private investment in precision agriculture.
- The program has now been **extended for another five years till Dec 2026**. Three components AIBP, Har Khet ko Pani and Watershed development has been extended.

- **How effective has the program been?**

- **Positives:**

- » The PMKSY has increased the coverage of irrigation to some extent.
- » It has also been able to increase the coverage of micro-irrigation and improving water use efficiency.
- » Further, in convergence with initiatives like Jal Shakti Abhiyan, the program has also contributed to watershed development and water conservation.

- **Limitations:**

- » There has been uneven progress across states, delays in fund release and weak convergence with allied schemes.
- » The adoption rate of micro-irrigation is still very low.
- » The "Har Khet ko Pani" component has also led to over-exploitation of groundwater.

- **Conclusion**

- The recent extension of various components of PMKSY till 2026 highlights that there is still a lot that needs to be done on the front of irrigation for Indian Agriculture. If PMKSY is able to achieve its targets by 2026, it would reduce vulnerability of Indian Agri sector to a great extent.

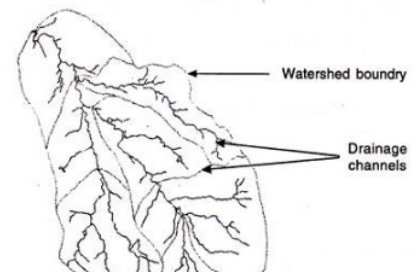
4) WATERSHED AND ASSOCIATED ISSUES

- **Questions**

- » Elaborate on the impact of the National Watershed Project in increasing agriculture production from water stressed areas [Mains 2019, 10 marks, 150 words]
- » What is watershed management? Discuss the key challenges that Watershed Management initiatives are facing in India. [12.5 marks, 200 words]

- **Introduction**

- » **Watershed** is an area of land where surface water drains down to a single point (stream, lake or ocean).
- » **Watershed development/management** is the treatment of the entire catchment area around the village to ensure conservation and regeneration of natural resources, especially water and its judicious use.
- » Watershed development is all about making running water stop and standing water to sink inside. It is the only option for rainfed areas for water conservation and recharge and to prevent soil degradation.



- **How is watershed development done?**

- » It is done from ridge to valley basis. Watershed development starts from the top-most point (ridge) and progresses downwards towards the valley.
 - Structures such as Water absorption trenches (WAT), Continuous contour trenches (CCT), stone bunds, check dams, percolation dams, ponds, and channels are built from the ridge to the valley.
- » **Main objective is to slow the movement of water**

- **Watershed Development Programs in India: Key Programs**
 - » **Integrated Watershed Management Program** (now a component of PMKSY)
 - » **National Watershed Development Project for Rainfed Areas (NWDPA)**
 - » **Watershed Development Project in Shifting Cultivation Area (WDPSCA)**
 - » **Watershed Development Fund (WDF)**
 - With the help of NABARD
 - Unified fund to help various watershed development programs in India.

- **How has watershed management programs contributed/ What are the objectives of these program?**
 - » **Water Conservation**
 - Increase in groundwater recharge.
 - Increase in number of wells and waterbodies (thus making water available during non-rainy season)
 - » **Control Floods**
 - » **Soil Conservation**
 - Increase in soil quality because of lesser erosion (thus protection of rich topsoil)
 - Reduces the rate of desertification.
 - Controlled problem of drainage, alkalinity, salinity etc.
 - » **Rehabilitate degraded lands**, restore productivity & hydrological functions of degraded lands
 - » **Better agricultural practices**
 - Enhancement of cropping intensity, diversity etc.
 - Changes in cropping pattern
 - Higher yield of crops
 - » **Integration of non-agricultural activities**
 - Increased availability of fodder promotes animal husbandry
 - » **Higher income for farmers**
 - Better socio-economic indicators
 - » **Capacity building of farmers**
 - To ensure sustainability

- **Some Limitations of the Programs**
 - » **Poor Implementation/ Delay**
 - **Parliamentary Standing Committee report submitted in July 2018 said that only 10% of the projects sanctioned between 2009-2015 have been completed.**
 - **Reluctance by the states to provide resources.**
 - **Why?** Less visibility
 - **Change in funding pattern (from a 90:10 Centre-State distribution to 60:40) has also contributed to the slow down.**
 - **Lack of coordination between different ministries**
 - Tendencies for our ministries to work in Silos have also become a factor hindering the growth of these projects
 - » **Lack of participation/encouragement** by all the stakeholders -> thus the program didn't show a lot of penetration.
 - » **Sustainability concerns**

- For instance, when the water table increases, farmers shift to water intensive crops like rice and sugarcane and thus deplete the water again. Thus, this misuse is not letting the projects become sustainable.
- There is a need to strengthen PRIs else the benefits will only be cyclical and short term.
- » **Lack of focus on capacity building**
 - This is further leading to less impact in long run as after the program construction period, the locals are not able to manage it properly.

- **Way Forward**

- » **Increased coordination** between Centre and State as well as among different ministries.
- » **Compulsorily involve locals/villagers** in the project to ensure better penetration of the watershed management skills and long-term sustainability.
- » **Increased funding and recognition** for states which take more efforts towards watershed management activities.
- » **Improved monitoring** of the project implementation through GIS, Space based technologies etc.
- » **Promote sustainability:** Farmers should be encouraged to avoid water intensive crops in dry and rainfed areas. Income support through assured prices for millets/pulses and other such less water consuming crops have to increase.

- **Conclusion:** Watershed development/management makes rainwater a nurturing, nourishing force of life, peace and prosperity. Taking into the consideration its extreme significance in increasing welfare of the region and fighting climate change, all the required efforts should be taken to ensure successful implementation of various ongoing projects.

5) MICRO-IRRIGATION

- **Questions**

- » How and to what extent micro-irrigation help in solving India's water crisis [Mains 2021, 10 marks, 150 words]
- » "Micro irrigation is an innovative technology to accelerate sustainable agriculture growth" Critically Analyze. [12.5 marks, 200 words]

- **Introduction**

- » Water is a scarce natural resource and faces a huge demand supply gap all over the world. Situation is worse in India which accommodates 17% of the world's population with 4% of the water resources.
- » In India, more than 80% of water is used for irrigation purposes. Therefore, optimal utilization of water in irrigation can play crucial role in ending the water scarcity that exists. Here, micro-irrigation can play a crucial role.

- **Micro Irrigation** is an innovative water saving technology in which water is directly supplied to crops with very less conveyance and evaporation losses. Main types of micro-irrigation system include:

- » **Drip Irrigation:** It allows water to drip slowly to the roots of plants, either from above the soil surface or buried below the surface. Crops cultivated under drip irrigation includes sugarcane, banana, cotton, lemon, grapes, oranges, mangoes, vegetables etc.

- » **Sprinkler Irrigation:** It uses water sprinklers to irrigate agri-crops. The water is applied in a controlled manner in a method that is similar to rainfall. Crops cultivated under sprinkler irrigation include, wheat, mustard, millet, sorghum etc
- » **Micro-Sprinkler Irrigation** provide irrigation with very fine droplets. They are suitable for low volume irrigation in horticulture crops, fruit flowers, greenhouses, nurseries etc.
- » **Porous Pipe System:** It is a system of sub-surface irrigation which can work on low pressure that can be provided by gravity (overhead tanks). Water is dispensed gradually near the roots of the plant/tree.
- » **Rain Gun:** Pressurized water through rain-guns is used to irrigate the crop.
- Drip Irrigation and Sprinkler Irrigation is the most common type of micro-irrigation system used in India.
- **Benefits of Micro Irrigation**
 - » **Better Water Use Efficiency:** Micro-Irrigation helps in significant reduction in losses through runoffs, evaporation losses and seepage losses. It can ensure higher water use efficiency up to 50-90%.
 - » **Fertilizer Use Efficiency:** Proper mixing of fertilizers and water, control of dosage, and direct application of fertilizers to the root zone result in saving of fertilizer consumption.
 - » **Energy Efficiency:** Micro irrigation requires minimal pressure and low flow rate. This reduces energy consumption. Even gravity can be used for this kind of irrigation. Off-grid farmers can utilize solar pumps for the purpose.
 - » **Increased Productivity:** Micro-irrigation is associated with higher crop yield which ensures good economic return for farmers.
 - » **Cost Saving** due to less water and fertilizer requirements. Further, the weeding effort reduces and thus the labor requirement goes down.
 - » **Crop Diversification:** Since water intensive crops can also be cultivated with less amount of water, it provides an option of going for diverse variety of crops.
 - » **Increased Farmer's income:** Reduction in input cost, increase in output and chances of growing diverse crops (including cash crops) increases farmer's income.
 - » **Drip Irrigation with mulching** can provide benefits like reduced evaporation, moisture control, reduced weed growth, reduced erosion etc.
- **Efforts towards promoting Micro-Irrigation in India**
 - » **National Mission on Micro Irrigation Program** which was later subsumed under **National Mission on Sustainable Agriculture** and is now being implemented under PMKSY-PDMC.
 - Under it various **Training and Awareness Programs, workshops, seminars and interactive meets** are conducted.
 - **50% subsidy** is provided to farmers for installing micro-irrigation system (MIS). (40% Centre, 10% state)
 - » **Micro-Irrigation Fund with NABARD** under PMKSY (May 2018)
 - It was set up in 2018 with the major objective of funding the states to facilitate them to mobilize resources which can be used to incentivize farmers towards micro-irrigation beyond the provisions available under PMKSY-PDMC.
 - » **NGOs such as Aga Khan Rural Support Program (AKRSP)** in Gujarat have played important roles in promoting micro-irrigation in the country.

- **Key Factors hindering the promotion of Micro-Irrigation in India:**
 - » **Energy Crisis** due to power outages and unscheduled interruptions across rural India
 - » **Initial Cost of Micro-Irrigation Infrastructure**
 - » **Declining Landholding and farm incomes**
- **Some Criticisms**
 - » Some experts feel that 'More Crop Per Drop' through micro-irrigation is a myth. In fact, earlier methods of irrigation with proper watershed management could be beneficial.
- **Way Forward:**
 - » **Integrate Drip Irrigation with solar panel in poor electricity areas:** This will reduce dependency on electricity and help efficient irrigation mechanisms.
 - » **Increases subsidy to small and marginal farmers** for installing micro-irrigation infrastructure. Promoting land pooling and pooling of micro-irrigation infrastructure.
 - » **More Research** is required to conclusively show that micro-irrigation reduces overall water use efficiency. Government should promote **more R&D** in new technologies in micro-irrigation system manufacturing. This may help in reducing the initial infrastructure cost required.
 - » **Promoting shift away from water intensive crops** in water scarce regions should also be promoted to reduce water scarcity.
- **Conclusion:** Micro irrigation has the potential to solve India's water scarcity crisis and political will to take above steps may contribute immensely towards accelerating sustainable agricultural growth.

6) ALLELOPATHY AND AGRICULTURE

- **Question:** What is allelopathy? Discuss its role in major cropping systems of irrigated agriculture [Mains 2016, 12.5 marks, 200 words]
- **Introduction:** Allelopathy is a biological phenomenon by which an **organism produces one or more biochemicals** that influence germination, growth, survival, and reproduction of other organisms. These biochemicals are known as allelochemicals and can be released in air, water or soil. These may have beneficial or detrimental effects on target organisms and the community.
- **It can play significant role in major cropping systems of irrigated agriculture:**
 - » **Increase nutrient availability** which can improve crop yield.
 - For e.g., wheatgrass can produce allelochemicals that can improve the quality of crops.
 - » **Sustainable Agriculture:** Allelopathy can contribute to sustainable agriculture by reducing the need for chemical inputs (like herbicides) and enhancing biodiversity.
 - » **Controlling Weeds:** For e.g., crop residue of rye when used as a cover crop in a no-till system, it releases allelochemicals and prevent growth of weeds.
 - » **Pest and Disease Control:** For e.g., Marigolds produce several allelochemicals in their roots and leaves. They are often planted in gardens and agricultural fields to deter various pests.
 - » **Making Crop Rotation and Intercropping better:** Understanding of allelopathy can help us understand which crops will benefits each other in case of crop rotation or inter-cropping.

- » **Soil Health:** Allelochemicals can influence the soil microbial community, affecting nutrient cycling, and soil fertility. They may either enhance or reduce soil health, depending on the type of allelochemicals involved.
- **However**, it has to be understood that lack of understanding of allelopathy can lead to several negative impacts. For e.g., Negative allelopathy can reduce crop yield if crops are grown in sequence or in combination with plants that produce inhibitory chemicals. They may inhibit germination, growth or reproduction.
 - » For e.g. Mustard produces a chemical called allyl isothiocyanate, which can inhibit the growth of some plants, such as tomatoes. Peanuts produce a chemical called gossypol, which can inhibit the growth of plants like corn.
- **Way Forward:**
 - » **More R&D** -> To understand allelopathic properties of various species and how to best utilize them.
 - » **Education and Extension:** By understanding allelopathy and the plants that can affect major cropping systems in irrigated agriculture, farmers can make informed decisions about crop rotations, cover crops, and other management practices to improve crop yields and reduce the use of pesticides.
 - » **Rotate crops** so that plants which have negative allelopathic impact are not planted in the same field year after year.
 - » **Commercialization:** There is a huge scope in development of products like bio-herbicides, from allelopathic plants. These could provide more suitable alternatives to synthetic agro-chemicals.
 - » **Policy Support:** Farmers should be provided financial incentives to adopt allelopathic practices, funding research in this area, and developing guidelines for the safe and effective use of allelopathic plants.
- **Conclusion:** Understanding of allelopathy in agriculture, particularly in irrigated cropping systems, can lead to development of more suitable farming practices. However, further research is needed to fully harness its potential and reduces the scope of negative impact.